Prescribed Decimal Prefixes and Prime Signatures: Existence and Minimum Formulas for OEIS A086558

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Abstract

For every $n \geq 1$ there is an n-digit integer beginning with the decimal digits of n and having the same prime signature as n. This proves the nonvanishing conjecture in OEIS A086558. We give a finite formula for the minimum using ordered prime tuples and one next-prime operation. At every prime-power index $n = p^e$ this simplifies to the e-th power of the first prime above an explicit integer root. The existence proof uses Dusart's explicit prime-interval theorem and fifteen small prime certificates.

1 The target interval

Let d be the decimal length of n, and put

$$L = n10^{n-d}, \qquad U = (n+1)10^{n-d}. \tag{1}$$

The integers in $[L, U)$ are exactly the n-digit integers beginning with n. A prime signature is the multiset of positive exponents in the prime factorization; the signature of 1 is empty. OEIS A086558 [1] takes the least integer in this interval with the signature of n, assigning zero if the set is empty.

Theorem 1. *For every $n \geq 1$ this set is nonempty.*

Proof. The case $n = 1$ has witness 1. For $n \geq 17$, let p be the largest prime factor of n, let $e = v_p(n)$, and put $C = n/p^e$. Define

$$x = (L/C)^{1/e} = p10^{(n-d)/e}, \qquad y = (U/C)^{1/e} = x(1 + 1/n)^{1/e}.$$

We will show that $x > 396738$ and

$$25 \log^2 x > e(n+1). \tag{2}$$

Dusart's Proposition 6.8 [2] then supplies a prime

$$x < q \leq x\left(1 + \frac{1}{25\log^2 x}\right) < y,$$

where the last inequality follows from

$$(1 + 1/n)^{1/e} - 1 \geq \frac{1}{e(n+1)}.$$

Indeed the derivative of $(1+t)^{1/e}$ on $[0, 1/n]$ is at least $1/(e(1+1/n))$, and integration proves the inequality. Since $q > x > p$, the prime q does not divide C. Therefore Cq^e lies in $[L, U)$ and has the same signature as n.

For $n \geq 64$, use $d \leq n/4$, $e \leq \log_2 n$, and the fact that $n/\log_2 n$ increases for $n > \mathrm{e}$. They give

$$x \geq 10^{3n/(4\log_2 n)} \geq 10^8 > 396738.$$

Also $(\log_2 n)^3 \leq 4n$ for $n \geq 32$: the ratio decreases for $\log n > 3$ and is $125/32 < 4$ at $n = 32$. Hence $e^3 \leq 4n$. Using $d \leq n/2$ and $\log 10 > 2$, we obtain

$$25\log^2 x > \frac{25n^2}{e^2} \geq \frac{25}{4}ne > e(n+1),$$

which proves (2). The elementary inequality $d \leq n/4$ for $n \geq 64$ follows on each interval of fixed digit length, starting with $d = 2$ and $n = 64$; for $d \geq 3$, $4d \leq 10^{d-1}$.

For $17 \leq n \leq 63$, $e \leq 5$. If $e = 1, 2, 3, 4, 5$, respectively, the smallest possible n in this range is at least $17, 18, 24, 48, 32$. Throughout the range $d = 2$, so $(n-d)/e \geq 6$ in every case. Thus $x \geq 10^6$ and $25\log^2 x > 25 \cdot 12^2 > 5 \cdot 64 \geq e(n+1)$.

The remaining cases $2 \leq n \leq 16$ have witnesses Cq^e in the table below; the fifteen displayed q are primes, verified by exhaustive trial division in the supplied program. Each displayed product has the required signature and decimal prefix.

n	C	q	e	n	C	q	e
2	1	23	1	10	5	200000033	1
3	1	307	1	11	1	11000000021	1
4	1	67	2	12	3	200003	2
5	1	50021	1	13	1	1300000000021	1
6	3	200003	1	14	7	2000000000003	1
7	1	7000003	1	15	5	30000000000011	1
8	1	431	3	16	1	6329	4
9	1	30011	2				

□

2 An exact finite prime-parameter formula

For an integer $z \geq 1$, let $\rho_e(z)$ be the least integer w with $w^e \geq z$, and let nextprime(z) be the least prime strictly greater than the integer z. All roots below are integer roots, including at exact power boundaries.

Theorem 2 (Minimum formula)**.** *Let $n \geq 2$ have exponent multiset $\{e_1, \ldots, e_t\}$. For each distinct ordering of these exponents, and each increasing tuple of primes $p_1 < \cdots < p_{t-1} < U$, put*

$$C = \prod_{i=1}^{t-1} p_i^{e_i}, \qquad q = \text{nextprime}\left(\max\left\{p_{t-1}, \rho_{e_t}(\lceil L/C \rceil) - 1\right\}\right).$$

When $t = 1$, take $C = 1$ and $p_0 = 1$. Form all candidates $Cq^{e_t} < U$. Then A086558(n) is their minimum. The tuple enumeration can be restricted by the explicit integer bounds

$$\left(\prod_{i<j} p_i^{e_i}\right) p_j^{e_j+\cdots+e_t} < U \qquad (1 \leq j < t). \tag{3}$$

Proof. Every candidate belongs to $[L, U)$ and has the prescribed signature, because q is distinct from the earlier primes and is the first prime meeting both the ordering and lower-value constraints. Conversely, factor a genuine minimum as $p_1^{e_1} \cdots p_t^{e_t}$ with $p_1 < \cdots < p_t$. Its exponent ordering is included, its first $t-1$ primes obey (3), and the formula's final prime q is at most p_t. The corresponding candidate is no larger than the genuine minimum, so equality follows. Theorem 1 ensures a candidate exists, and every tuple range is finite. □

Corollary 1 (Closed prime-power-index reduction)**.** *If $n = p^e$ is a prime power, then*

$$\text{A086558}(n) = \left[\text{nextprime}\left(\rho_e(n10^{n-d}) - 1\right)\right]^e. \tag{4}$$

For $n = 1$ the value is 1.

For example, at $n = 16$, $L = 16 \cdot 10^{14}$ and the first eligible prime fourth root is 6329, giving $6329^4 = 1604502443050081$. Formula (4) follows directly by ordering prime powers. The existence theorem guarantees its candidate remains below U. For $n \geq 17$ this also follows from the prime interval in the proof; the prime-power cases below 17 are certified in the table.

The formula uses finite minimization over prime tuples; it is not a constant-cost expression in elementary operations. Its evaluator and independent trial-division certificates use only the Python standard library. Initial formula outputs are compared with the published values.

References

[1] A. Murthy, OEIS A086558, https://oeis.org/A086558, accessed October 26, 2026.

[2] P. Dusart, *Estimates of some functions over primes without R.H.*, arXiv:1002.0442v1 (2010), Proposition 6.8, https://arxiv.org/abs/1002.0442v1.